

Commercial-in-confidence

Annual pricing report 2007–08 Electricity Distribution Services

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ActewAGL

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1. Introduction

The Independent Competition and Regulatory Commission (the Commission), as the jurisdictional regulator for the ACT, is responsible for determining the annual revenue requirement for services provided by ActewAGL's electricity distribution network. The principles under which this revenue requirement is determined are outlined in the National Electricity Rules (the Rules). Translation of the revenue requirement into network pricing was provided for under part E of chapter 6 of the National Electricity Code, the rules applying at the time of the Commission's pricing determination. The Code allowed the Commission, in consultation with Registered Participants, to develop an alternative pricing methodology.

The Commission decided to use an alternative pricing methodology for the regulatory control period commencing on 1 July 2004. As part of that methodology, the Commission requires ActewAGL to provide the following documents explaining charges for prescribed distribution services:

- a pricing strategy statement. The pricing strategy statement for the period 2004/05 to 2008/09 was provided to the Commission together with ActewAGL's annual pricing statement for 2004/05 in May 2004; and
- an annual pricing statement. The annual pricing statement for 2007/08 is this document.

The Commission requires that the annual pricing reports contain:

- a demonstration of compliance with the determined revenue cap (the Commission has indicated that it will consider this section of the report commercial in confidence and will not require it to be made publicly available);
- a list of new prices;
- a discussion of price changes (in excess of 5% higher than the natural price change allowed by the Commission's determination) or changes in the structure of prices, and any new or obsolete tariffs;
- a demonstration of the impact on typical customers' bills and forecast average prices (based on typical bill categories as set by the Energy Supply Association of Australia);
- an explanation of how these prices comply with the medium term pricing strategy, or any change in the strategy (substantial change to the strategy requires a new pricing strategy to be approved by the Commission);
- an outline of the expected levels of service and projected capital expenditure programs for the year;
- an outline of the expected consumption for major customer classes and maximum demand for the year.

Once the Commission has approved prices for the coming financial year, this document is to be disclosed publicly on ActewAGL's and the Commission's websites together with the pricing strategy statement already published.

2. Regulated revenue cap

2.1 Prices consistent with the regulated revenue cap

The Commission regulates ActewAGL's electricity distribution charges through its Price Direction for the five-year period from 1 July 2004.¹

The Commission has prescribed that the price of distribution services associated with the conveyance of electricity through the distribution system, including miscellaneous monopoly services, are subject to economic regulation and regulated using a maximum average revenue cap. The Commission excluded metering services from the revenue cap for prescribed distribution services and regulates metering charges for consumers using less than 160 MWh per annum under a separate revenue cap for such excluded distribution services (see Section 7).

2.2 Base maximum average revenue for prescribed services

ActewAGL's distribution prices are regulated using a maximum average revenue cap. In its Final Decision of March 2004, the Commission determined that the maximum average revenue for 2003/04 to be \$0.0383 per kWh.²

To calculate the maximum average revenue for 2007/08, the following formula is used:

$$\text{Maximum average revenue per kWh} = (1 + (\text{CPI}_t + X_t)) \times (\text{MAR}_{t-1})$$

where:

- MAR_{t-1} is the maximum average revenue per kWh in the previous year; and
- CPI means the Consumer Price Index, All Groups Index number for the weighted average of eight capital cities.

The CPI figure will be determined using the following formula:

$$\text{CPI}_{(t)} = \frac{\text{CPI}_{\text{March}(t-2)} + \text{CPI}_{\text{June}(t-2)} + \text{CPI}_{\text{Sept}(t-1)} + \text{CPI}_{\text{Dec}(t-1)}}{\text{CPI}_{\text{March}(t-3)} + \text{CPI}_{\text{June}(t-3)} + \text{CPI}_{\text{Sept}(t-2)} + \text{CPI}_{\text{Dec}(t-2)}} - 1$$

where:

- year t is the year for which tariffs are being set; and
- year $t - 1$ is the previous regulatory year.

The CPI adjustment figure for the year commencing 1 July 2007 is calculated as follows:

¹ ICRC 2004, *Final Decision – Investigation into prices for electricity distribution services in the ACT*, Appendix 2.

² ICRC 2004, p xxv.

$$CPI_{(2007)} = \frac{CPI_{March(2006)} + CPI_{June(2006)} + CPI_{Sept(2006)} + CPI_{Dec(2006)}}{CPI_{March(2005)} + CPI_{June(2005)} + CPI_{Sept(2005)} + CPI_{Dec(2005)}} - 1$$

Therefore:

$$CPI_{(2007)} = \frac{151.9 + 154.3 + 155.7 + 155.5}{147.5 + 148.4 + 149.8 + 150.6} - 1$$

$$CPI_{(2007)} = \frac{617.4}{596.3} - 1$$

$$CPI_{(2007)} = 3.54\%$$

The Commission has determined the X factor for 2007/08 to be zero. The CPI is applied to the maximum average revenue for 2006/07 to obtain the maximum average revenue in 2007/08. The calculations of the maximum average revenue are shown in Table 2.1.

Table 2.1 Calculation of the Maximum Average Revenue 2007/08

	Base MAR	X Factor	CPI	MAR
2004/05	\$0.0383	6.80%	2.77%	\$0.03676
2005/06	\$0.03676	0%	2.34%	\$0.03762
2006/07	\$0.03762	0%	2.67%	\$0.03862
2007/08	\$0.03862	0%	3.54%	\$0.03999

The maximum average revenue cap is applied to the actual energy sales in the previous calendar year³ to establish a price cap for the following financial year. Therefore, the prices for 2007/08 are based upon sales in 2006.

2.3 Pass-through events

The Commission has provided for ActewAGL to seek to incorporate in proposed tariffs additional costs incurred in connection with prescribed pass-through events.⁴ There are two events relevant to this submission:

- the ACT Government has introduced a new tax — the Utilities Network Facilities Tax (UNFT) — that ActewAGL has proposed to recover by passing through the cost of the tax in network charges; and
- ActewAGL had identified a need to pass through costs of the proposed Southern Electricity Supply Point.

³ ICRC 2004, Appendix 2. Part 3 (p 132).

⁴ ICRC 2004, Appendix 2. Part 6 (p 136).

2.3.1 Utilities network facilities tax

The *Utilities (Network Facilities Tax) Act 2006* came into effect on 21 December 2006, implementing the UNFT from 1 January 2007. In April 2007, the Commission amended its March 2004 Final Decision to enable it to allow the pass-through of costs incurred as a result of state and territory taxes.⁵

It is proposed that ActewAGL Distribution recover in the 2007/08 network prices the full amount forecast in the budget estimates for the first 18 months. If any adjustment were necessary to reflect the actual amount of the tax billed to ActewAGL for the 18 month period, it would be made in the 2008/09 network charges.

2.3.2 Southern Electricity Supply Point

On 21 August 2006, ActewAGL provided a submission to the Commission seeking approval for the basis and process for passing through the efficient costs associated with the establishment of a southern electricity supply point to the ACT. This submission foreshadowed formal pass through claims to be included in ActewAGL's annual pricing proposals in March 2007 and 2008.

In a letter of 18 October 2006, the Commission agreed to consider this matter in consultation with the Australian Energy Regulator (AER) at the time of ActewAGL's pricing submission for 2007-08.

However, due to increasing uncertainty over the timing of cost forecasts resulting from land access and development approval issues, ActewAGL now proposes that these costs be passed through on a calendar year basis once actual costs are known (rather than be based on forecast financial year costs as originally proposed). This means ActewAGL is not proposing a cost pass-through in 2007-08 prices, but rather that actual 2007 calendar year costs passed through in 2008-09 prices. Planning and design costs already incurred would be rolled into the regulatory asset base at 1 July 2009.

2.4 Calculation of the maximum revenue cap

The total revenue cap for prescribed distribution services is made up of:

- the base revenue cap for prescribed distribution services calculated using the maximum average revenue cap; and
- the pass-through of the utilities network facilities tax.

The revenue to be recovered from distribution use of system (DUOS) charges is the remainder after deducting revenue from miscellaneous fees.

⁵ ICRC 2007, Final Decision Electricity Distribution Services: Proposed Amendment to the 2004 Pricing Direction.

3. Distribution use of system charges

ActewAGL has prepared distribution use of system prices for 2007/08 as shown in Table 3.1. Note in Table 3.1 that ActewAGL is introducing a new residential time-of-use network charge, a new low voltage capacity network charge principally for embedded generators and has restructured the charges for un-metered loads.

The residential time-of-use network charge is being proposed as an option for consumers with interval meters arising from the Commission's decision of 19 December 2005.⁶ The Commission's price determination in 2004 provides that:

If a new tariff is introduced to replace an existing tariff the commission will use a forecast based on the estimated demand in the year before the new tariff is intended to come into effect.⁷

The new tariff does not replace any existing tariff. Nor does ActewAGL anticipate a significant move to the residential time-of-use network charge in the first year. Hence it has not made a forecast of the demand on this tariff. Consumers with a relatively higher off-peak load are those most likely to move to this charge to lower their network bill. Therefore, ActewAGL Distribution is not anticipating any additional revenue from consumers moving to this new charge.

The new low voltage capacity network charge is available to commercial consumers as an optional network charge. Low voltage commercial consumers with generators connected directly to their electricity network will be required to use this network charge. Existing commercial customers will opt for this new tariff only if it reduces their network costs. Therefore, ActewAGL Distribution is not anticipating any additional revenue from this tariff.

The restructure of prices for small un-metered loads is to simplify the prices and make them more equitable. The restructured tariff is to apply to the existing load.

⁶ ICRC, Final Decision – Review of Metrology Procedures

⁷ ICRC, Final Decision – Investigation into prices for electricity distribution services in the ACT, p132.

Table 3.1 Distribution use of system charges 2007/08

<i>Customer Class</i>	<i>2006/07 DUOS Prices (Excluding GST)</i>	<i>2007/08 DUOS Prices (Excluding GST)</i>
Residential Standard Network		
Network access charge (c/day)	12.50	14.30
Energy (c/kWh)	3.40	3.70
Total		
Residential 5,000 Network		
Network access charge (c/day)	32.5	34.30
Energy (c/kWh)		
To 60 kWh per day	1.8	2.10
Greater than 60kWh per day	3.4	3.70
Total		
Residential Heat Pump Network		
Network access charge (c/day)	75.5	77.30
Energy (c/kWh)		
To 165 kWh per day	0.3	0.60
Greater than 165 kWh per day	3.4	3.70
Total		
Residential TOU Network		
Network access charge (c/day)		14.30
Max (c/kWh)		5.10
Mid (c/kWh)		3.50
Economy (c/kWh)		2.60
Controlled economy (c/kWh)		2.60
Total		
Solar Energy	0.00	0.00
Off-Peak (1/2) Night Network (c/kWh)	0.10	0.32
Off-Peak (3) Day & Night Network (c/kWh)	0.50	0.90
General Network		
Network access charge (c/day)	24.10	26.00
Energy (c/kWh)		
To 330 kWh per day	5.57	6.00
Greater than 330 kWh per day	7.43	7.76
Total		
General TOU Network		
Network access charge (c/day)	25.00	26.00
Business (c/kWh)	8.00	8.65
Evening (c/kWh)	4.45	4.70
Off-Peak (c/kWh)	2.30	2.60
Total		

<i>Customer Class</i>	<i>2006/07 DUOS Prices (Excluding GST)</i>	<i>2007/08 DUOS Prices (Excluding GST)</i>
LV TOU Demand Network		
Network access charge (\$/day)	0.25	0.36
Business (c/kWh)	1.75	1.90
Evening (c/kWh)	1.70	1.60
Off-Peak (c/kWh)	1.05	1.25
Maximum Demand (c/kVA/day)	26.60	28.60
or (c/KW/day)	44.30	48.30
Total		
LV TOU Capacity Network		
Network access charge (\$/day)		0.36
Business (c/kWh)		1.90
Evening (c/kWh)		1.60
Off-Peak (c/kWh)		1.20
Maximum Demand (c/kVA/day)		13.50
Capacity Charge (c/kVA/day)		13.50
Total		
HV TOU Demand Network - Customer LV		
Network access charge (\$/day)	12.00	12.00
Business (c/kWh)	1.50	1.40
Evening (c/kWh)	1.43	1.30
Off-Peak (c/kWh)	0.90	1.10
Maximum Demand (c/kVA/day)	10.00	11.00
Capacity Charge (c/kVA/day)	10.00	11.00
Total		
HV TOU Demand Network		
Network access charge (\$/day)	12.00	12.00
Business (c/kWh)	1.75	1.65
Evening (c/kWh)	1.65	1.50
Off-Peak (c/kWh)	1.00	1.20
Maximum Demand (c/kVA/day)	10.00	11.00
Capacity Charge (c/kVA/day)	10.00	11.00
Total		
Streetlight Network		
Network access charge (c/day)	24.10	26.00
Energy (c/kWh)	3.18	3.48
Total		

Small Unmetered Loads Network

Network access charge (c/day)

Small Unmetered loads with assessed ratings of:

50 Watts or less 12.0

more than 50 but less than 150W 24.0

more than 150 but less than 300W 48.0

Supply charge (c/W per day)

more than 300 W but less than 1kW 0.157

Network access charge (c/day/NMI) 26.00

Energy (c/kWh) 8.00

Total

Internal Network

Network access charge (c/day) 4.00 4.00

Energy (c/kWh) 0.21 0.33

Total

4. Miscellaneous service charges

ActewAGL proposes charges for prescribed miscellaneous services as shown in Table 4.1.

The *prescribed services* revenue from these charges, when applied to units sold in 2006, would have recovered \$1,100,362. Metering is an excluded distribution service. The revenue related to metering charges is regulated under a separate revenue cap.

In line with ActewAGL's Pricing Strategy Statement, the schedule includes a new service charge of \$136.36 (\$150 including GST) for the installation of an interval meter at a consumer's request.

Table 4.1 Proposed miscellaneous fees and charges

Service charges	Price 2007/08 Excluding GST
For visit to re-energise or de-energise a premises	
• Business Hours - re-energise	\$43.64
• After Hours - re-energise	\$94.55
• Business Hours - de-energise	\$40.00
Meter test during business hours (refundable deposit)	\$54.55
De-energise premises for non-payment	\$73.64
Special meter reading, for example check read (second in billing period)	\$28.18
Install interval meter at customer's request.	\$136.36
Temporary Connections	
• Overhead	\$313.64
• Standard underground	\$454.55
• Free standing underground	\$554.55
Overhead: Remove, reposition or disconnect service, per site visit	\$227.27
Underground: Remove, reposition or disconnect service, per site visit	\$545.45
Upgrade service from single to three phase at customer's request where load does not justify 3 phase (S&I Rules clause 3.10)	
• Overhead	\$290.91
• Underground - Service Cable not required	\$290.91
• Underground - Service Cable Replacement required	\$545.45
Installation defect (where a revisit to a site is necessitated by obstructed access or non-compliance with S&I Rules)	\$150.00
Issue of copies of electrical drawings (per sheet or electronic equivalent)	\$11.82
De-energise wires (to allow safer approach, eg, for tree pruning, plant operation, oversize loads, construction activities)	\$363.64

5. Transmission use of system charges

The Australian Competition and Consumer Commission (ACCC) regulates transmission use of system (TUOS) charges. TUOS charges are not part of the regulated revenue requirement. The Commission, in accordance with the Rules, “allows ActewAGL to recover reasonable costs arising from, but not limited to, charges paid to transmission network service providers arising out of the provision of distribution services”.⁸

TransGrid is yet to publish its transmission charges for NSW and the ACT for 2007/08. However, informal correspondence with TransGrid has indicated that TUOS prices will rise by about 5.0% plus CPI in 2007/08.

The Commission’s determination requires ActewAGL’s distribution prices to be calculated on actual sales in 2006. ActewAGL has estimated the TUOS cost for network pricing purposes by applying the forecast TUOS prices for 2007/08 to the actual energy imported into the ACT in the 2006 calendar year.

Table 5.1 shows the network charges that ActewAGL proposes to apply to recover these costs.

Table 5.1 TUOS charges 2007/08

<i>Customer Class</i>	<i>2007/08 TUOS Charges (Excluding GST)</i>
Residential Standard Network	
Energy (c/kWh)	0.50
Residential 5,000 Network	
Energy (c/kWh)	
To 60 kWh per day	0.50
Greater than 60kWh per day	0.50
Residential Heat Pump Network	
Energy (c/kWh)	
To 165 kWh per day	0.50
Greater than 165 kWh per day	0.50
Residential TOU Network	
Max (c/kWh)	1.00
Mid (c/kWh)	0.50
Economy (c/kWh)	0.00
Controlled economy (c/kWh)	0.00
Off-Peak (1/2) Night Network (c/kWh)	0.10
Off-Peak (3) Day & Night Network (c/kWh)	0.15

⁸ ICRC 2004, *Final Decision – Investigation into prices for electricity distribution services in the ACT*, p119

<i>Customer Class</i>	<i>2007/08 TUOS Charges (Excluding GST)</i>
General Network	
Energy (c/kWh)	
To 330 kWh per day	0.80
Greater than 330 kWh per day	0.84
General TOU Network	
Business (c/kWh)	2.00
Evening (c/kWh)	1.00
Off-peak (c/kWh)	0.00
LV TOU Demand Network	
Business (c/kWh)	0.75
Evening (c/kWh)	0.55
Off-peak (c/kWh)	0.00
Maximum Demand (c/kVA/day)	3.40
or (c/KW/day)	5.70
LV TOU Capacity Network	
Business (c/kWh)	0.75
Evening (c/kWh)	0.55
Off-peak (c/kWh)	0.00
Maximum Demand (c/kVA/day)	1.50
Capacity Charge (c/kVA/day)	1.50
HV TOU Demand Network - Customer LV	
Business (c/kWh)	0.70
Evening (c/kWh)	0.50
Off-peak (c/kWh)	0.00
Maximum Demand (c/kVA/day)	2.00
Capacity Charge (c/kVA/day)	2.00
HV TOU Demand Network	
Business (c/kWh)	0.70
Evening (c/kWh)	0.50
Off-peak (c/kWh)	0.00
Maximum Demand (c/kVA/day)	2.00
Capacity Charge (c/kVA/day)	2.00
Street-lighting Network	
Energy (c/kWh)	0.52
Small Unmetered Loads Network	
Energy (c/kWh)	0.50
Internal Network	
Energy (c/kWh)	0.23

6. Network use of system charges

The proposed network use of system (NUOS) charges for 2007/08 comprise the distribution use of system (DUOS) charges and TUOS charges. The NUOS charges are shown in Table 6.1.

Table 6.1 Network use of system charges 2007/08 (excluding GST)

Customer Class	2006/07 Network Prices	2007/08 DUOS Prices	2007/08 TUOS charges	2007/08 Network Prices
Residential Standard Network				
Network access charge (c/day)	12.50	14.30	0.00	14.30
Energy (c/kWh)	3.90	3.70	0.50	4.20
Residential 5,000 Network				
Network access charge (c/day)	32.50	34.30	0.00	34.30
Energy (c/kWh)				
To 60 kWh per day	2.30	2.10	0.50	2.60
Greater than 60kWh per day	3.90	3.70	0.50	4.20
Residential Heat Pump Network				
Network access charge (c/day)	75.50	77.30	0.00	77.30
Energy (c/kWh)				
To 165 kWh per day	0.80	0.60	0.50	1.10
Greater than 165 kWh per day	3.90	3.70	0.50	4.20
Residential TOU Network				
Network access charge (c/day)		14.30	0.00	14.30
Max (c/kWh)		5.10	1.00	6.10
Mid (c/kWh)		3.50	0.50	4.00
Economy (c/kWh)		2.60	0.00	2.60
Controlled economy (c/kWh)		2.60	0.00	2.60
Solar Energy	0.00	0.00	0.00	0.00
Off-Peak (1/2) Night Network (c/kWh)	0.20	0.32	0.10	0.42
Off-Peak (3) Day & Night Network (c/kWh)	0.80	0.90	0.15	1.05
General Network				
Network access charge (c/day)	24.10	26.00	0.00	26.00
Energy (c/kWh)				
To 330 kWh per day	6.37	6.00	0.80	6.80
Greater than 330 kWh per day	8.27	7.76	0.84	8.60
General TOU Network				
Network access charge (c/day)	25.00	26.00	0.00	26.00
Business (c/kWh)	10.00	8.65	2.00	10.65
Evening (c/kWh)	5.45	4.70	1.00	5.70
Off-Peak (c/kWh)	2.30	2.60	0.00	2.60

<i>Customer Class</i>	<i>2006/07 Network Prices</i>	<i>2007/08 DUOS Prices</i>	<i>2007/08 TUOS charges</i>	<i>2007/08 Network Prices</i>
LV TOU Demand Network				
Network access charge (\$/day)	0.25	0.36	0.00	0.36
Business (c/kWh)	2.50	1.90	0.75	2.65
Evening (c/kWh)	2.25	1.60	0.55	2.15
Off-Peak (c/kWh)	1.05	1.25	0.00	1.25
Maximum Demand (c/kVA/day)	30.00	28.60	3.40	32.00
or (c/KW/day)	50.00	48.30	5.70	54.00
LV TOU Capacity Network				
Network access charge (\$/day)		0.36	0.00	0.36
Business (c/kWh)		1.90	0.75	2.65
Evening (c/kWh)		1.60	0.55	2.15
Off-Peak (c/kWh)		1.25	0.00	1.25
Maximum Demand (c/kVA/day)		13.50	1.50	15.00
Capacity Charge (c/kVA/day)		13.50	1.50	15.00
HV TOU Demand Network - Customer LV				
Network access charge (\$/day)	12.00	12.00	0.00	12.00
Business (c/kWh)	2.20	1.40	0.70	2.10
Evening (c/kWh)	1.93	1.30	0.50	1.80
Off-Peak (c/kWh)	0.90	1.10	0.00	1.10
Maximum Demand (c/kVA/day)	12.00	11.00	2.00	13.00
Capacity Charge (c/kVA/day)	12.00	11.00	2.00	13.00
HV TOU Demand Network				
Network access charge (\$/day)	12.00	12.00	0.00	12.00
Business (c/kWh)	2.45	1.65	0.70	2.35
Evening (c/kWh)	2.15	1.50	0.50	2.00
Off-Peak (c/kWh)	1.00	1.20	0.00	1.20
Maximum Demand (c/kVA/day)	12.00	11.00	2.00	13.00
Capacity Charge (c/kVA/day)	12.00	11.00	2.00	13.00
Street-lighting Network				
Supply Charge (c/day)	24.10	26.00	0.00	26.00
Energy (c/kWh)	3.70	3.48	0.52	4.00
Small Unmetered Loads Network				
Network access charge (c/day)				
Small Unmetered loads with assessed ratings of:				
50 Watts or less	13.00			
more than 50 but less than 150 W	26.00			
more than 150 but less than 300 W	52.00			
Supply charge (c/W per day)				
more than 300 W but less than 1 kW	0.170			
Network access charge (c/day)		26.00	0.00	26.00
Energy (c/kWh)		8.00	0.50	8.50
Internal Network				
Network access charge (c/day)	4.00	4.00	0.00	4.00
Energy (c/kWh)	0.50	0.33	0.23	0.56

7. Excluded Services

7.1 Regulated metering service charges

The Commission capped revenue from metering for ActewAGL customers consuming less than 160 MWh at \$5.09 million in 2004/05. The amount is to be adjusted for inflation over the regulatory period. In 2006/07, the cap was \$5,348,189. In 2006, the inflation rate was 3.54% raising the revenue cap for metering services for 2007/08 to \$5,537,515. This cap applies also to revenue from miscellaneous fees and charges relating to metering.

In its Review of Metrology Procedures,⁹ the Commission states that it does not wish ActewAGL to be disadvantaged by its decision to introduce interval meters¹⁰ and allowed initial costs to be recovered in 2006/07 prices. ActewAGL has now employed a metering officer to manage the introduction of interval meters. Costs incurred due to the introduction of interval meters up until 31 December 2006 have amounted to \$50,437. It is proposed to recover this cost in the metering charges for 2007/08. This takes the total metering revenue requirement to \$5,587,952.

The proposed metering charges and the forecast revenue from those charges are shown in Table 7.1.

Table 7.1 Proposed metering charges and forecast revenue 2007/08

	2006/07 Price \$ per day (excluding GST)	2007/08 Price \$ per day (excluding GST)
Residential accumulation meters ¹¹	\$0.094	\$0.097
Residential interval meters (read quarterly as an interval meter)		\$0.48
Business accumulation meters	\$0.164	\$0.17
Internal accumulation meters	\$0.051	\$0.05
Business time of use meters (obsolete)	\$0.75	\$0.17
Business interval meters	\$1.75	\$1.64
Special meter reading	\$27.27	\$25.18
Meter test (deposit)	\$52.73	\$54.55
Install interval meter (by request)		\$136.36
TOTAL		

By default, the new residential interval meters will be read quarterly as time-of-use meters. Where consumers or retailers require interval metering data to be read, ActewAGL has proposed a new metering charge to recover the additional metering cost. For simplicity, where

⁹ ICRC 2006, *Final decision: Review of metrology procedures*, p28

¹⁰ ICRC 2006, p11

¹¹ All residential meters other than interval meters read as interval meters.

three-phase interval meters are read quarterly, they will have the same metering charge as single-phase interval meters read quarterly.

From 1 July 2007, interval meters of consumers using more than 100 MWh pa are required to be read as interval meters.¹² Consumers on the business time-of-use tariff using less than 100 MWh will have a reduction in their metering charge. This is because all new meters will be interval meters and all interval meters for consumers using less than 100 MWh pa will be read by default as time-of-use meters. This default arrangement will be applied to existing customers with time-of-use meters who will pay the standard meter reading charge.

ActewAGL has about 50 customers on the business time-of-use tariff using more than 100 MWh pa. It has installed its standard interval meters in these premises to minimise the number of meter types required to be stocked. To avoid the anomaly that a substantial increase in metering charges to these customers would create, ActewAGL proposes to continue to read the time-of-use data from these meters and charge the customers accordingly.

ActewAGL Distribution will be installing interval meters in all new installations and as replacement meters. Where a consumer requests to exchange their existing accumulation meter for an interval meter, ActewAGL will apply a charge of \$150 (including GST) to recover the labour cost of around \$135 (excluding GST) to remove an existing meter and install a new meter.

Additional costs of interval meters will be passed through in future years as the expense is incurred.

¹² ICRC, *Final Decision: Review of Metrology Procedures*, p31.

8. Price changes

8.1 Network prices excluding metering

Network charges for 2007/08, *excluding* metering, are shown in Table 8.1.

Table 8.1 Network charges 2007/08 (excluding metering)

Customer Class	2007/08 Network Charges	
	(excluding GST)	(including GST)
Residential Standard Network		
Network access charge (c/day)	14.30	15.73
Energy (c/kWh)	4.20	4.62
Residential 5,000 Network		
Network access charge (c/day)	34.30	37.73
Energy (c/kWh)		
To 60 kWh per day	2.60	2.86
Greater than 60kWh per day	4.20	4.62
Residential Heat Pump Network		
Network access charge (c/day)	77.30	85.03
Energy (c/kWh)		
To 165 kWh per day	1.10	1.21
Greater than 165 kWh per day	4.20	4.62
Residential TOU Network		
Network access charge (c/day)	14.30	15.73
Max (c/kWh)	6.10	6.71
Mid (c/kWh)	4.00	4.40
Economy (c/kWh)	2.60	2.86
Controlled economy (c/kWh)	2.60	2.86
Solar Energy	0.00	0.00
Off-Peak (1/2) Night Network (c/kWh)	0.42	0.462
Off-Peak (3) Day & Night Network (c/kWh)	1.05	1.155
General Network		
Network access charge (c/day)	26.00	28.60
Energy (c/kWh)		
To 330 kWh per day	6.80	7.48
Greater than 330 kWh per day	8.60	9.46
General TOU Network		
Network access charge (c/day)	26.00	28.60
Business (c/kWh)	10.65	11.715
Evening (c/kWh)	5.70	6.27
Off-Peak (c/kWh)	2.60	2.86

Customer Class	2007/08 Network Charges	
	(excluding GST)	(including GST)
LV TOU Demand Network		
Network access charge (\$/day)	0.36	0.396
Business (c/kWh)	2.65	2.915
Evening (c/kWh)	2.15	2.365
Off-Peak (c/kWh)	1.25	1.375
Maximum Demand (c/kVA/day)	32.00	35.20
or (c/KW/day)	54.00	59.40
LV TOU Capacity Network		
Network access charge (\$/day)	0.36	0.396
Business (c/kWh)	2.65	2.915
Evening (c/kWh)	2.15	2.365
Off-Peak (c/kWh)	1.25	1.375
Maximum Demand (c/kVA/day)	15.00	16.50
Capacity Charge (c/kVA/day)	15.00	16.50
HV TOU Demand Network – Customer LV		
Network access charge (\$/day)	12.00	13.20
Business (c/kWh)	2.10	2.31
Evening (c/kWh)	1.80	1.98
Off-Peak (c/kWh)	1.10	1.21
Maximum Demand (c/kVA/day)	13.00	14.30
Capacity Charge (c/kVA/day)	13.00	14.30
HV TOU Demand Network		
Network access charge (\$/day)	12.00	13.20
Business (c/kWh)	2.35	2.585
Evening (c/kWh)	2.00	2.20
Off-Peak (c/kWh)	1.20	1.32
Maximum Demand (c/kVA/day)	13.00	14.30
Capacity Charge (c/kVA/day)	13.00	14.30
Streetlighting Network		
Network access charge (c/day)	26.00	28.60
Energy (c/kWh)	4.00	4.40
Small Unmetered Loads Network		
Network access charge (c/day)	26.00	28.60
Energy (c/kWh)	8.50	9.35
Internal Network		
Network access charge (c/day)	4.00	4.40
Energy (c/kWh)	0.56	0.616

8.2 Change in metering charges

The change in metering charges in 2007/08 for customer with loads less than 160 MWh per annum is shown in Table 8.2.

Table 8.2 Proposed metering charges 2007/08

	2006/07 Price per day (excluding GST)	2007/08 Price per day (excluding GST)	Change %
Residential accumulation meters	\$0.095	\$0.097	2.1%
Residential interval meters (read quarterly as an interval meter)		\$0.48	
Business accumulation meters	\$0.169	\$0.170	0.6%
Internal accumulation meters	\$0.050	\$0.050	0.0%
Business time of use meters	\$0.75	\$0.170	-77.3%
Business interval meters	\$1.75	\$1.64	-6.3%
Special meter reading	\$25.00	\$28.18	3.3%
Meter test (deposit)	\$50.00	\$54.55	3.5%
Install interval meter (by request)		\$136.36	

As discussed in Section 7.1, there are several changes to metering charges related to the introduction of interval metering.

There is a new residential interval metering charge for residential interval meters read as interval meters. This charge recovers the higher cost associated with reading these meters and processing the data. These meters have been programmed by default to provide time-of-use data. If only the time-of-use data is read, the meter charge will be the same as for accumulation meters.

The introduction of interval meters has made the charge for business time-of-use meters obsolete. All new meters will be interval meters and they will be read, by default, as time-of-use meters and be charged at the rate for accumulation meters. There is no reason to distinguish between old meters read as time-of-use meters and new (interval) meters read as time-of-use meters and, therefore, no longer any basis for separate charges for time-of-use meters. Hence the discontinuation of this charge results in the 77.3% reduction in the metering charge for business time of use meters.

ActewAGL will be installing interval meters as replacement meters and in all new installations. Interval meters are also available to customers on request. If customers make such a request, they will be adding further to the cost of interval meters. That cost must be recovered in metering charges. If there were no direct cost to the consumer, some might unnecessarily request the new meter. To avoid such outcomes and make charges more cost reflective, ActewAGL proposes that where a consumer requests an interval meter be installed to replace an existing meter, a charge of \$150 (including GST) be applied.

8.3 Network prices including metering

For comparative purposes, prices *including* metering are listed in Table 8.3 with the rate of increase for each component and the total for the average customer, where appropriate. The average increase in prices from 2004/05 to 2007/08 is also shown.

Table 8.3 Network charges 2007/08 (including metering, excluding GST)

Customer Class	2004/05 Network Charges	2006/07 Network Charges	2007/08 Network Charges	04/05 to 07/08 Average annual Price Change	06/07 to 07/08 Price Change
Residential Standard Network					
Network access charge (c/day)	20.70	22.00	24.00	5.1%	9.1%
Energy (c/kWh)	3.90	3.90	4.20	2.5%	7.7%
Net change for average consumer				3.1%	8.0%
Residential 5,000 Network					
Network access charge (c/day)	41.00	42.00	44.00	2.4%	4.8%
Energy (c/kWh)					
To 60 kWh per day	2.30	2.30	2.60	4.2%	13.0%
Greater than 60kWh per day	3.90	3.90	4.20	2.5%	7.7%
Net change for average consumer				3.5%	10.2%
Residential Heat Pump Network					
Network access charge (c/day)	84.00	85.00	87.00	1.2%	2.4%
Energy (c/kWh)					
To 165 kWh per day	0.80	0.80	1.10	11.2%	37.5%
Greater than 165 kWh per day	3.90	3.90	4.20	2.5%	7.7%
Net change for average consumer				4.5%	13.3%
Residential TOU Network					
Network access charge (c/day)			24.00		
Max (c/kWh)			6.10		
Mid (c/kWh)			4.00		
Economy (c/kWh)			2.60		
Controlled economy (c/kWh)			2.60		
Solar Energy	0.00	0.00	0.00	0.0%	0.0%
Off-Peak (1/2) Night Network (c/kWh)	0.18	0.20	0.42	32.6%	110.0%
Off-Peak (3) Day & Night Network (c/kWh)	0.80	0.80	1.05	9.5%	31.3%
General Network					
Network access charge (c/day)	40.00	41.00	43.00	2.4%	4.9%
Energy (c/kWh)					
To 330 kWh per day	6.35	6.37	6.80	2.3%	6.8%
Greater than 330 kWh per day	8.25	8.27	8.60	1.4%	4.0%
Net change for average consumer				2.2%	6.3%

Customer Class	2004/05 Network Charges	2006/07 Network Charges	2007/08 Network Charges	04/05 to 07/08 Average annual Price Change	06/07 to 07/08 Price Change
General TOU Network					
Network access charge (c/day)	95.00	100.00	43.00	-23.2%	-57.0%
Business (c/kWh)	9.95	10.00	10.65	2.3%	6.5%
Evening (c/kWh)	5.45	5.45	5.70	1.5%	4.6%
Off-Peak (c/kWh)	2.30	2.30	2.60	4.2%	13.0%
Net change for average consumer				1.7%	4.6%
LV TOU Demand Network					
Network Access (\$/day)	2.00	2.00	2.00	0.0%	0.0%
Business (c/kWh)	2.40	2.50	2.65	3.4%	6.0%
Evening (c/kWh)	2.20	2.25	2.15	-0.8%	-4.4%
Off-Peak (c/kWh)	1.00	1.05	1.25	7.7%	19.0%
Maximum Demand (c/kVA/day)	30.00	30.00	32.00	2.2%	6.7%
or (c/KW/day)	50.00	50.00	54.00	2.6%	8.0%
Net change for average consumer				2.7%	6.9%
LV TOU Capacity Network					
Network Access (\$/day)			2.00		
Business (c/kWh)			2.65		
Evening (c/kWh)			2.15		
Off-Peak (c/kWh)			1.25		
Maximum Demand (c/kVA/day)			15.00		
Capacity Charge (c/kVA/day)			15.00		
HV TOU Demand Network - Customer LV					
Network Access (\$/day)	13.71	13.75	13.64	-0.2%	-0.8%
Business (c/kWh)	2.10	2.20	2.10	0.0%	-4.5%
Evening (c/kWh)	1.89	1.93	1.80	-1.6%	-6.7%
Off-Peak (c/kWh)	0.85	0.90	1.10	9.0%	22.2%
Maximum Demand (c/kVA/day)	12.00	12.00	13.00	2.7%	8.3%
Capacity Charge (c/kVA/day)	12.00	12.00	13.00	2.7%	8.3%
Net change for average consumer				2.6%	5.7%
HV TOU Demand Network					
Network Access (\$/day)	13.71	13.75	13.64	-0.2%	-0.8%
Business (c/kWh)	2.35	2.45	2.35	0.0%	-4.1%
Evening (c/kWh)	2.10	2.15	2.00	-1.6%	-7.0%
Off-Peak (c/kWh)	0.98	1.00	1.20	7.0%	20.0%
Maximum Demand (c/kVA/day)	12.00	12.00	13.00	2.7%	8.3%
Capacity Charge (c/kVA/day)	12.00	12.00	13.00	2.7%	8.3%
Net change for average consumer				2.3%	5.4%
Streetlighting Network					
Network access charge (c/day)	40.00	41.00	43.00	2.4%	4.9%
Energy (c/kWh)	3.65	3.70	4.00	3.1%	8.1%
Net change for average consumer				3.1%	8.1%

<i>Customer Class</i>	<i>2004/05 Network Charges</i>	<i>2006/07 Network Charges</i>	<i>2007/08 Network Charges</i>	<i>04/05 to 07/08 Average annual Price Change</i>	<i>06/07 to 07/08 Price Change</i>
Small Unmetered Loads Network					
Network access charge (c/day)					
Small Unmetered loads with assessed ratings of:					
50 Watts or less	13.00	13.00			
more than 50 but less than 150 W	26.00	26.00			
more than 150 but less than 300 W	52.00	52.00			
Network access charge (c/W per day)					
more than 300 W but less than 1 kW	0.17	0.17			
Network access charge (c/day)			26.00		
Energy (c/kWh)			8.50		
Internal Network					
Network Access (c/day)	9.00	9.00	9.00	0.0%	0.0%
Energy (c/kWh)	0.50	0.50	0.56	3.8%	12.0%
Net change for average consumer				2.9%	8.9%
Net change for average consumer				2.8%	7.3%

9. Discussion of price changes

9.1 Network use of system charges

The Commission's alternative pricing methodology requires "a discussion of price changes (in excess of 5% higher than the natural price change¹³ allowed by the commission's determination) or changes in the structure of prices, and any new or obsolete tariffs".¹⁴ Given a CPI of 3.54%, and an X factor of 0, ActewAGL is required to explain price increases 5% above that level; that is, greater than 8.54%.

The network charges in 2007/08 include pass-through costs averaging 0.2 cents per kWh for the costs of the new Utilities Network Facilities Tax. It is the pass-through of this cost that fully explains all but two instances in Table 8.3 where the increase in network use of system charges is greater than 8.54%. The exceptions are the *residential heat pump network charge* and the *off-peak (1/2) night network charge*.

The charge per kWh to 165 kWh per day for the *residential heat pump network charge* will increase from 0.8 cents to 1.1 cents, an increase of 0.3 cents. After deducting the 0.2 cents per kWh, the 0.1 cents per kWh increase is an increase of 12.5%. The amount of the increase is the same as for the *Residential standard* and *Residential 5,000* network charges. It is necessary to keep the changes in the *Residential network* charges approximately the same in nominal terms. The relative increase is large because the energy charge is very low. This tariff has a relatively high fixed charge that increased 2.4%. The average increase for consumers on this tariff is 5.5%, after excluding the impact of the 0.2 cents per kWh due to the tax.

The *off-peak (1/2) night network charge* will increase from 0.20 cents to 0.42 cents per kWh, an increase of 0.22 cents. After deducting the 0.20 cents per kWh passed through for the tax, the 0.02 cents per kWh increase is a 10.0% price rise. This increase of two hundredths of a cent is extremely small but the relative increase is large because the network charge is extremely low.

¹³ That is, the CPI plus X increase allowed by the Commission's final price determination.

¹⁴ ICRC 2004, Appendix 6 clause 5.2.

9.2 New network use of system charges

ActewAGL Distribution has introduced a new residential time-of-use network charge. This charge complements the Commission's requirement to install residential interval meters in new installations and as replacements for existing residential meters. Consumers will be able to request interval meters be installed to replace existing accumulation meters.

The residential time-of-use network charge is made up of an access charge and the following energy charges:

- **Max**, for energy consumed daily between 7.00 am and 9.00 am and between 5.00 pm and 8.00 pm;
- **Mid**, for energy consumed daily between 9.00 am and 5.00 pm and between 8.00 pm and 10.00 pm;
- **Economy**, for energy consumed each day between 10.00 pm and 7.00 am the following day; and
- **Controlled Economy**, for energy recorded on the controlled load element for appliances such as water heaters that make energy available for limited periods during the economy and mid time periods.

The default meter readings for customers with interval meters and consuming less than 100 MWh per annum will provide *Max*, *Mid* and *Economy* accumulation consumption data. If the consumer's meter has a controlled load element (for off-peak appliances), their readings would include *Controlled Economy* consumption accumulation data. Residential customers with interval meters will have access to the residential time-of-use network charge regardless of whether or not they have their meter read as an interval meter.

Residential consumers that have their meter read as an interval meter will not have access to the *Controlled economy* rate (or any off-peak tariff) as the meter is programmed to produce only a single profile of consumption data i.e., there is not a separate half hour consumption profile for the controlled load element. Even so, consumers may continue to have their off-peak controlled load appliances automatically switched to come on at off-peak times.

Consumers with interval meters will have access to this new tariff. However, it is not necessary for a residential consumer with interval metering data to use the residential time-of-use network charge. All other residential network charges continue to be available to customers with interval meters.

Consumers will only move to this charge if they perceive it to better meet their needs. ActewAGL cannot expect to earn any additional revenue from a charge where customers on the whole receive cost savings.

9.3 Embedded generation for commercial sites

ActewAGL receives enquiries from time to time regarding small embedded generation units (less than 5 MW) and is in the process of reviewing its policy on generation and associated pricing issues to better meet customer requirements, and those of ActewAGL and the National Electricity Market.

Customers that have a generator on their side of the meter are able to reduce network costs. Yet ActewAGL is required to continue to provide sufficient back-up capacity for when the generator is not operating. To reflect the cost of providing this facility, ActewAGL proposes that low voltage consumers with embedded generation be required to use the proposed low voltage capacity network charge. This charge has a capacity component that is applied to the maximum demand over the previous 12 months. This charge provides the consumer with an incentive to operate the generator so as to best manage their demand on the distribution network and ensures that they make an equitable contribution towards the cost of operating the network.

9.4 Small un-metered loads

It is proposed to change the structure of the network charge for small un-metered loads to introduce a fixed charge per NMI and a variable energy charge based upon the estimated energy consumption. The current charges classify un-metered loads into four categories:

- 50 watts or less;
- more than 50 watts but not more than 150 watts;
- more than 150 watts but not more than 300 watts; and
- more than 300 watts but not more than 1kW.

These charges mean that a load of 10 watts pays the same as a load five times the size and a load of 55 watts pays twice the price of a 50 watt load. The new charges improve equity between such loads.

Although these consumers are not metered, NEMMCO allocates a national meter identifier (NMI) number for such consumers. There are 18 NMI's for un-metered loads in the ACT. It is proposed to charge a fixed daily charge for each NMI and an energy charge for the estimated energy consumption, as for other tariffs.

The new tariffs and the structural change are consistent with ActewAGL's pricing strategy, which is primarily that its distribution charges be cost reflective. The introduction of interval meters will provide time-of-use consumption data that enables ActewAGL to provide customers with the most appropriate pricing signals to inform their behaviour so as to attain efficient utilisation and expansion of the network. The low voltage capacity network charge will reflect the cost to ActewAGL of providing back-up capacity to a consumer with an embedded generator. The structural change in the charges for un-metered loads removes the block

charges and applies a charge that more accurately reflects the costs of these loads to the network.

9.5 Miscellaneous service charges

Table 9.1 shows the proposed miscellaneous service charges for 2007/08 together with the percentage change. There are no increases greater than 8.54%. However, there is a new charge for installing interval meters at a customer's request. This new charge is discussed in Sections 7.1 and 8.2.

This charge of \$150 (including GST) reflects the additional cost to ActewAGL of installing the new meter. It does not recover any of the cost of the meter nor the cost of the redundant pre-existing meter but it is expected that a charge would discourage unjustified requests for interval meters that could needlessly raise metering costs for other consumers.

Table 9.1 Changes to miscellaneous service charges

<i>Service charges</i>	<i>Price 2006/07 Including GST</i>	<i>Price 2007/08 Including GST</i>	<i>Change</i>
For visit to re-energise or de-energise a premises			
• Business Hours - re-energise	\$47.00	\$48.00	2.1%
• After Hours - re-energise	\$100.00	\$104.00	4.0%
• Business Hours - de-energise	\$42.00	\$44.00	4.8%
Meter test during business hours (refundable deposit- conditions)	\$58.00	\$60.00	3.4%
De-energise premises for non-payment	\$78.00	\$81.00	3.8%
Special meter reading, for example check read (second in billing period)	\$30.00	\$31.00	3.3%
Install interval meter at customers request		\$150.00	n.a.
Temporary Connections			
• Overhead	\$336.00	\$345.00	2.7%
• Standard underground	\$483.00	\$500.00	3.5%
• Free standing underground	\$588.00	\$610.00	3.7%
Overhead: Remove, reposition or disconnect service, per site visit	\$242.00	\$250.00	3.3%
Underground: Remove, reposition or disconnect service, per site visit	\$575.00	\$600.00	4.3%
Upgrade service from single to three phase at customer's request where load does not justify 3 phase (S&I Rules clause 3.10)			
• Overhead	\$310.00	\$320.00	3.2%
• Underground - Service Cable not required	\$310.00	\$320.00	3.2%
• Underground - Service Cable Replacement required	\$575.00	\$600.00	4.3%
Installation defect (applied where a revisit to a site is necessitated by obstructed access or non-compliance with Service and Installation Rules.)	\$158.00	\$165.00	4.4%
Issue of copies of electrical drawings (per sheet or electronic equivalent)	\$12.50	\$13.00	4.0%
De-energise wires (to allow safer approach, eg, for tree pruning, plant operation, oversize loads, construction activities)	\$395.00	\$400.00	1.3%

9.6 Impact on consumer bills

The Commission requires that ActewAGL demonstrate the impact on typical customers' bills and forecast average prices (based on typical bill categories as set by the Energy Supply Association of Australia (ESAA)) of its proposed tariff. This is subject to the comment in section 2.4 above regarding the status of this price proposal.

9.6.1 Domestic network electricity price

ActewAGL has four domestic electricity network charges:

- *Residential Standard* network charge;
- *Residential 5,000* network charge;
- *Residential with Heat Pump* network charge; and the new
- Residential time-of-use network charge.

Customers are expected to move to the residential time-of-use charge only if there are savings to be made. There is no existing residential time-of-use charge to compare this charge to. The impact on typical domestic customers' bills for the other three charges based on typical bill categories as set by the ESAA is shown in Table 9.2. These bills include metering.

Table 9.2 Impact on typical domestic customers' forecast bills

Network Charge Name	Typical Domestic Network Electricity Bills including GST							
	Consumption - kWh per year							
	Year	600	1,200	2,500	3,500	5,000	7,500	10,000
Residential Standard	2006/07	\$114.07	\$139.81	\$195.58	\$238.48	\$302.83	\$410.08	\$517.33
	2007/08	\$124.08	\$151.80	\$211.86	\$258.06	\$327.36	\$442.86	\$558.36
	Change %	8.8%	8.6%	8.3%	8.2%	8.1%	8.0%	7.9%
Residential 5,000	2006/07	\$183.81	\$198.99	\$231.88	\$257.18	\$295.13	\$358.38	\$421.63
	2007/08	\$193.82	\$210.98	\$248.16	\$276.76	\$319.66	\$391.16	\$462.66
	Change %	5.4%	6.0%	7.0%	7.6%	8.3%	9.1%	9.7%
Residential with Heat Pump	2006/07	\$346.56	\$351.84	\$363.28	\$372.08	\$385.28	\$407.28	\$429.28
	2007/08	\$356.57	\$363.83	\$379.56	\$391.66	\$409.81	\$440.06	\$470.31
	Change %	2.9%	3.4%	4.5%	5.3%	6.4%	8.0%	9.6%

The impact on typical domestic customers' forecast average network prices, including metering, is shown in Table 9.3.

Table 9.3 Impact on typical domestic customers' forecast network prices

Network Charge Name	Typical Domestic Network Forecast Average Prices							
	Consumption - kWh per year							
	Year	600	1,200	2,500	3,500	5,000	7,500	10,000
Residential Standard	2006/07	19.01	11.65	7.82	6.81	6.06	5.47	5.17
	2007/08	20.68	12.65	8.47	7.37	6.55	5.90	5.58
	Change %	8.8%	8.6%	8.3%	8.2%	8.1%	8.0%	7.9%
Residential 5,000	2006/07	30.64	16.58	9.28	7.35	5.90	4.78	4.22
	2007/08	32.30	17.58	9.93	7.91	6.39	5.22	4.63
	Change %	5.4%	6.0%	7.0%	7.6%	8.3%	9.1%	9.7%
Residential with Heat Pump	2006/07	57.76	29.32	14.53	10.63	7.71	5.43	4.29
	2007/08	59.43	30.32	15.18	11.19	8.20	5.87	4.70
	Change %	2.9%	3.4%	4.5%	5.3%	6.4%	8.0%	9.6%

9.6.2 Domestic off-peak in conjunction with domestic network charge

ActewAGL offers three domestic off-peak network charges:

- *Off-peak (1) Night* (obsolete);
- *Off-peak (2) Night* (obsolete and at the same rate as off-peak (1) night); and
- *Off-Peak (3) Day & Night*.

These charges are available to domestic consumers on the *Residential Standard Network* charge. The impact on typical customers' bills of the proposed network charges for the off-peak based on typical bill categories as set by the ESAA is shown in Table 9.4.

Table 9.4 Impact on typical domestic off-peak customers' network bills

Network Charge Name	Typical Domestic Network Electricity Bills with Off-peak including GST Consumption - kWh per year						
	Aggregate	5,000	5,000	7,500	7,500	13,000	13,000
	Off-peak	1,300	2,500	2,000	5,000	3,900	6,500
Residential standard with off-peak (1/2) Night	2005/06	\$249.92	\$201.08	\$328.68	\$206.58	\$487.30	\$381.48
	2006/07	\$273.31	\$223.41	\$359.70	\$234.96	\$534.80	\$426.69
	Change %	9.4%	11.1%	9.4%	13.7%	9.7%	11.9%
Residential standard with off-peak (3) day & night	2005/06	\$258.50	\$217.58	\$341.88	\$239.58	\$513.04	\$424.38
	2006/07	\$282.32	\$240.74	\$373.56	\$269.61	\$561.83	\$471.74
	Change %	9.2%	10.6%	9.3%	12.5%	9.5%	11.2%

The impact on typical off-peak customers' forecast average network prices is shown in Table 9.5.

Table 9.5 Impact on typical domestic off-peak customers' average network prices

Network Charge Name	Typical Domestic Network Forecast Average Prices c/kWh including GST Consumption - kWh per year						
	Aggregate	5,000	5,000	7,500	7,500	13,000	13,000
	Off-peak	1,300	2,500	2,000	5,000	3,900	6,500
Residential standard with off-peak (1/2) Night	2006/07	5.00	4.02	4.38	2.75	3.75	2.93
	2007/08	5.47	4.47	4.80	3.13	4.11	3.28
	Change %	9.4%	11.1%	9.4%	13.7%	9.7%	11.9%
Residential standard with off-peak (3) day & night	2006/07	5.17	4.35	4.56	3.19	3.95	3.26
	2007/08	5.65	4.81	4.98	3.59	4.32	3.63
	Change %	9.2%	10.6%	9.3%	12.5%	9.5%	11.2%

9.6.3 Small business electricity prices

The impact on typical small business customers' bills of the proposed network charges for the general network charge based on typical bill categories as set by the ESAA is shown in Table 9.6.

Table 9.6 Impact on typical general network charge customers' network bills

Typical Small Business Network Electricity Bills General Network charge including metering, excluding GST Consumption - MWh per year						
Year	6	15	30	40	60	80
2006/07	\$531.85	\$1,105.15	\$2,060.65	\$2,697.65	\$3,971.65	\$5,245.65
2007/08	\$564.95	\$1,176.95	\$2,196.95	\$2,876.95	\$4,236.95	\$5,596.95
Change %	6.2%	6.5%	6.6%	6.6%	6.7%	6.7%

The impact on typical business customers' forecast average network prices is shown in Table 9.7.

Table 9.7 Impact on typical general network charge customers' average prices

Typical Average Small Business Network Electricity Prices General Network Charge including metering Consumption - MWh per year						
Year	6	15	30	40	60	80
	Average electricity price - c/kWh (GST exclusive)					
2006/07	8.86	7.37	6.87	6.74	6.62	6.56
2007/08	9.42	7.85	7.32	7.19	7.06	7.00
Change %	6.2%	6.5%	6.6%	6.6%	6.7%	6.7%

9.6.4 Small business time of use electricity prices

The impact on typical small business customers' bills of the proposed network charges for the general TOU network charge based on typical bill categories as set by the ESAA is shown in Table 9.8. Also shown are the prices using the average profile for consumers on this network tariff in the ACT. It should be noted that the reduction in the metering charge for this tariff class is likely to make this tariff attractive to many more small businesses, particularly those with a relatively high off-peak load.

Table 9.8 Impact on typical general TOU customers' network bills

Typical Small Business Network Electricity Bills General TOU Network charge including metering, excluding GST Consumption - MWh per year							
	Year	20	30	40	60	80	100
ESSA profile 25% off-peak	2006/07	\$1,752.50	\$2,446.25	\$3,140.00	\$4,527.50	\$5,915.00	\$7,302.50
	2007/08	\$1,636.95	\$2,376.95	\$3,116.95	\$4,596.95	\$6,076.95	\$7,556.95
	Change %	-6.6%	-2.8%	-0.7%	1.5%	2.7%	3.5%
ActewAGL Profile 52% off-peak	2006/07	\$1,442.76	\$1,981.65	\$2,520.53	\$3,598.29	\$4,676.06	\$5,753.82
	2007/08	\$1,317.40	\$1,897.62	\$2,477.85	\$3,638.30	\$4,798.74	\$5,959.19
	Change %	-8.7%	-4.2%	-1.7%	1.1%	2.6%	3.6%

The impact on typical general TOU customers' forecast average network prices is shown in Table 9.9.

Table 9.9 Impact on typical general TOU customers' average network prices

Typical Small Business Network Electricity Prices General TOU Network Charge including metering Consumption - MWh per year							
	Year	20	30	40	60	80	100
Average electricity price - c/kWh (GST exclusive)							
ESSA profile	2006/07	8.76	8.15	7.85	7.55	7.39	7.30
25% off-peak	2007/08	8.18	7.92	7.79	7.66	7.60	7.56
	Change %	-6.6%	-2.8%	-0.7%	1.5%	2.7%	3.5%
ActewAGL profile	2006/07	7.21	6.61	6.30	6.00	5.85	5.75
52% off-peak	2007/08	6.59	6.33	6.19	6.06	6.00	5.96
	Change %	-8.7%	-4.2%	-1.7%	1.1%	2.6%	3.6%

9.6.5 Small business electricity prices for contestable customers

The impact on typical small contestable business customers' bills of the proposed network charges for the low voltage demand network charge based on typical bill categories as set by the ESAA is shown in Table 9.10. It is not possible to duplicate the ESAA profile to derive these prices. Instead of using 100 kW as the maximum demand, a maximum demand of 100 kVA was used. The same energy consumption was used as in the ESAA's analysis. The mix of business, evening and off-peak energy used was that for an average customer on the low voltage demand network charge.

Table 9.10 Impact on typical small business LV demand customers' network bills

Typical Small Business Network Electricity Bills for Contestable Customers Low Voltage Demand charge with annual Peak demand of 100 kVA Includes metering, excludes GST						
Year	Load Factor					
	20%	30%	40%	50%	60%	80%
	Consumption - MWh per year					
	175	263	350	438	526	701
2006/07	\$14,878.29	\$16,477.44	\$18,076.59	\$19,675.73	\$21,274.88	\$24,473.17
2007/08	\$15,847.35	\$17,566.03	\$19,284.71	\$21,003.39	\$22,722.06	\$26,159.42
Change %	6.5%	6.6%	6.7%	6.7%	6.8%	6.9%

The impact on typical small low voltage demand customers' forecast average network prices is shown in Table 9.11.

Table 9.11 Impact on small business LV demand customers' network prices

Typical Small Business Network Electricity Prices for Contestable Customers Low Voltage Demand charge with annual Peak demand of 100 kVA Consumption - MWh per year						
Year	Load Factor					
	20%	30%	40%	50%	60%	80%
	Consumption - MWh per year					
	175	263	350	438	526	701
	Average electricity price - c/kWh (GST exclusive)					
2006/07	8.49	6.27	5.16	4.49	4.05	3.49
2007/08	9.05	6.68	5.50	4.80	4.32	3.73
Change %	6.5%	6.6%	6.7%	6.7%	6.8%	6.9%

9.6.6 Medium size business electricity prices for contestable customers

The impact on typical medium size contestable business customers' bills of the proposed network charges for the low voltage demand network charge based on typical bill categories as set by the ESAA is shown in Table 9.12. As for the small consumers, it was not possible to duplicate the ESAA profile to derive these prices. Instead of using 250 kW as the maximum demand, a maximum demand of 250 kVA was used. The same energy consumption was used as in the ESAA's analysis. The mix of business, evening and off-peak energy used was that for an average customer on the low voltage demand network charge.

Table 9.12 Impact on medium size LV demand customers' network bills

Typical Medium Size Business Network Electricity Bills for Contestable Customers Low Voltage Demand charge with annual Peak demand of 250 kVA Includes metering, excludes GST						
Year	Load Factor					
	20%	30%	40%	50%	60%	80%
	Consumption - MWh per year					
	438	657	876	1,095	1,314	1,752
2006/07	\$36,100.73	\$40,098.60	\$44,096.46	\$48,094.33	\$52,092.20	\$60,087.93
2007/08	\$38,523.39	\$42,820.08	\$47,116.77	\$51,413.47	\$55,710.16	\$64,303.54
Change %	6.7%	6.8%	6.8%	6.9%	6.9%	7.0%

The impact on typical medium size low voltage demand customers' forecast average network prices is shown in Table 9.13. It is worth noting that while the rate of increase is lower for the consumer with the low load factor than the high low factor, the absolute increase in average prices is higher for consumers with the low load factor.

Table 9.13 Impact on medium size LV demand customers' network prices

Typical Medium Size Business Network Electricity Prices for Contestable Customers Low Voltage Demand charge with annual Peak demand of 250 kVA Consumption - MWh per year						
Year	Load Factor					
	20%	30%	40%	50%	60%	80%
	Consumption - MWh per year					
	438	657	876	1,095	1,314	1,752
	Average electricity price - c/kWh (GST exclusive)					
	8.24	6.10	5.03	4.39	3.96	3.43
2006/07	8.24	6.10	5.03	4.39	3.96	3.43
2007/08	8.80	6.52	5.38	4.70	4.24	3.67
Change %	6.7%	6.8%	6.8%	6.9%	6.9%	7.0%

9.6.7 Large size business electricity prices for contestable customers

The impact on typical large size contestable business customers' bills of the proposed network charges for the high voltage demand network charge based on typical bill categories as set by the ESAA is shown in Table 9.14.

As for the low voltage consumers, it was not possible to duplicate the ESAA profile to derive these prices. Instead of using 1,000 kW as the maximum demand, a maximum demand of 1,000 kVA was used and the capacity charge was applied to this as well as to the maximum demand charges. The same amount of energy consumption was used as in the ESAA's analysis. The mix of business, evening and off-peak energy used was that for an average customer on the high voltage demand network charge.

Table 9.14 Impact on large size HV demand customers' network bills

Typical Large Size Business Network Electricity Bills for Contestable Customers High Voltage Demand charge with annual Peak demand of 1,000 kVA Includes metering, excludes GST						
Year	Load Factor					
	20%	30%	40%	50%	60%	80%
	Consumption - MWh per year					
	1,752	2,628	3,504	4,380	5,256	7,008
2006/07	\$119,814.89	\$133,412.96	\$147,011.03	\$160,609.10	\$174,207.18	\$201,403.32
2007/08	\$127,710.53	\$141,626.50	\$155,542.46	\$169,458.43	\$183,374.40	\$211,206.33
Change %	6.6%	6.2%	5.8%	5.5%	5.3%	4.9%

The impact on typical large size high voltage demand customers' forecast average network prices is shown in Table 9.15.

Table 9.15 Impact on typical large size HV demand customers' average network prices

Typical Large Size Business Network Electricity Prices for Contestable Customers High Voltage Demand charge with annual Peak demand of 1,000 kVA Consumption - MWh per year						
Year	Load Factor					
	20%	30%	40%	50%	60%	80%
	Consumption - MWh per year					
	1,752	2,628	3,504	4,380	5,256	7,008
	Average electricity price - c/kWh (GST exclusive)					
2006/07	6.84	5.08	4.20	3.67	3.31	2.87
2007/08	7.29	5.39	4.44	3.87	3.49	3.01
Change %	6.6%	6.2%	5.8%	5.5%	5.3%	4.9%

9.7 Compliance with the medium term strategy

ActewAGL's medium term price strategy is to set network prices to be cost reflective so as to provide consumers with the most appropriate cost signals under the revenue cap. As explained in section 2.2 of ActewAGL's *Pricing Strategy Statement 2004/05 to 2008/09*, ActewAGL's network pricing methodology allocates costs to consumers to generate prices that recover the regulated network revenue. This approach has been continued in preparing its 2007/08 prices.

10. Distribution services

10.1 Expected level of service

ActewAGL's investment in, and maintenance of the network ensure that the current standard of service will be maintained in 2007/08 in accordance with customer requirements as derived from ActewAGL's willingness to pay study.¹⁵

10.2 Maximum demand

Maximum demand for the 2007/08 is forecast to be 596 MW and it is expected to be attained in the winter of 2007.

¹⁵ NERA & AC Nielsen 2003, *Willingness to Pay Research Study*..